

libutils: audit summary

Date: October 4, 2026 · **Sources:** `audit/2026-10-04-{dependencies,quality,tests,bugs}.md` , commit `2b46e19` + the working tree changes

How to read this document

- Summary: one row per audit with its verdict and link.
- Top actions across the audits, highest gain first.
- What was not run and why.

Each audit has its own report with the evidence; this page only aggregates them. Not legal advice for the license points.

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1. Summary

Audit	Verdict	High	Medium	Low	Report
Dependencies	clean: license scope stated (MIT for libutils only, CC BY-NC-SA 4.0 for the synced algorithms), OpenSSL 3.5.9 without advisory	0	0	1	dependencies
Quality	9.2 / 10, very consistent style, no warning in libutils' own code	0	1	4	quality
Tests coverage	854 / 854 pass, lines 88.99 %, functions 94.12 %, branches 74.01 %, 86 % of the public API	—	—	—	tests
Bugs / UB	no bug: ASan + UBSan clean, TSan / LSan reports explained false positives, threaded suites stable over 30 runs	0	0	0	bugs

2. Top actions

- Split `ArgParser::parseFlags_` and `Cli::getInput_` (~170 lines each). Gain: readability of the two most complex parsers. Effort: medium, covered by the tests. (*quality 3.1*)
- Fix upstream the 14 `c2dmp-hsm` conversion warnings and add an SPDX line in the sync `sed` step of `c2dmp-hsm / s.o.s`. Effort: small. (*quality 3.4, dependencies 4*)
- Run the `SharedMemory` client in a fork ed child in `ReadableLastOnly`. Gain: a clean TSan run. Effort: small. (*bugs 3*)
- Decide on the unreachable `autoCast` cases (`Settings.hpp:126-157`). Effort: small. (*tests 5*)
- Add error-code cases to raise the branch coverage (74 %), starting with `Cli.cpp` and `Settings.cpp`. Effort: medium. (*tests 5*)
- Clean the root (`sandbox/` for the scratch programs, delete `a.out / latex/`). Effort: minutes. (*quality 3.2*)

3. Not run

Item	Reason
<code>clang-tidy</code> , <code>cppcheck</code> , <code>scan-build</code>	not installed (<code>dnf install clang-tools-extra cppcheck clang-analyzer</code> , needs root)
<code>valgrind</code>	replaced by the ASan / LSan run
Optimized / Asan release builds	not built (Debug sanitizer builds only)

Item	Reason
Debian / Ubuntu dependency path (libssl3)	audit machine is Fedora 44

4. Audited commit

2b46e19 (Merge branch 'main' of github.com:TsukiNi22/libutils , local main) with the changes of the working tree; date October 4, 2026.